Aryan Naik

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PROFILE

Second-year MEng Robotics and Artificial Intelligence student at University College London with hands-on experience building production-grade AI agents, multi-agent pipelines, and autonomous systems. Seeking to apply machine learning engineering and data pipeline skills to industrial AI challenges in the energy sector, where reliable model deployment and human-in-the-loop oversight are critical to operational outcomes.

EDUCATION

MEng Robotics and Artificial Intelligence University College London (UCL), London, UK Expected Graduation: 2029

PROJECTS

Job Drop AI Agent Python, LangGraph, AWS Bedrock, Sentence-Transformers, SQLite, Telegram API, pdfkit

- Reduced manual job-tracking overhead across a full interview pipeline (applied, phone screen, technical, final, outcome) by engineering a multi-agent LangGraph state machine with graph checkpointing, enabling end-to-end application lifecycle management within a single auditable workflow.
- Improved candidate-to-role match quality by implementing a retrieval-augmented generation triage layer using sentence-transformers with cosine similarity scoring, enabling ranked filtering of scraped listings across Reed.co.uk and Gradcracker before any human review.
- Increased pipeline reliability and model evaluation quality by injecting historical interview outcome data as context at inference time, creating a self-improving feedback loop that conditions future agent decisions on past performance signals.
- Maintained human oversight of all AI-generated outputs by integrating a Telegram human-in-the-loop approval workflow with graph checkpointing, ensuring no CV or cover letter was submitted without explicit operator sign-off.

Aegis Swarm ROS 2 Jazzy, FastDDS, NVIDIA Isaac Sim, Unity C#, OpenXR, Meta Quest 3

- Reduced physical hardware risk during early-stage swarm development by building a distributed drone autonomy system validated entirely within an NVIDIA Isaac Sim digital twin, enabling iterative architecture testing across multiple agents before any real-world deployment.
- Achieved sub-100ms command round-trip latency in simulation by configuring a FastDDS middleware layer for ROS 2 Jazzy inter-node communication, meeting the real-time coordination threshold required for stable multi-agent swarm behaviour.
- Enabled single-operator teleoperation of a multi-drone swarm by implementing a Meta Quest 3 OpenXR interface in Unity C#, reducing the operator headcount required for manual swarm oversight from multiple users to one.

SKILLS

Languages: Python, C# **AI and ML:** LangGraph, multi-agent systems, retrieval-augmented generation, sentence-transformers, cosine similarity scoring, AWS Bedrock, model evaluation, self-improving pipelines, human-in-the-loop workflows **Data and MLOps:** SQLite, data pipelines, graph checkpointing, historical context injection, web scraping, pdfkit **Robotics and Simulation:** ROS 2, FastDDS, NVIDIA Isaac Sim, digital twin simulation, OpenXR **Industrial Relevance:** autonomous systems, operator teleoperation interfaces, real-time distributed coordination, production-grade agent deployment

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Hiring Team Schlumberger

Dear Hiring Team,

Schlumberger operates at the intersection of energy and technology, deploying intelligent systems across some of the most demanding environments on earth. The scale at which Schlumberger applies data-driven decision-making and autonomous tooling to real-world subsurface challenges is precisely the kind of problem space I want to contribute to. As a second-year MEng Robotics and Artificial Intelligence student at University College London, I am eager to bring my applied AI experience to your team this summer.

My most relevant project is Job Drop, an end-to-end multi-agent AI system I designed and built in Python using LangGraph and AWS Bedrock with Claude Sonnet as the inference backbone. I implemented a retrieval-augmented generation triage pipeline using sentence-transformers and cosine similarity scoring, automated multi-source web scraping across live job boards, and built a full SQLite-backed application tracking system covering every stage of the interview pipeline. The system improves over time by injecting historical interview context into future decisions, and incorporates a Human-in-the-Loop approval workflow via Telegram with graph checkpointing for reliability. Building this end-to-end taught me how to architect production-grade agentic systems that balance automation with human oversight at each critical decision point.

Joining Schlumberger as an AI Engineer Intern would allow me to apply these skills to genuinely high-stakes engineering problems, where the cost of a poorly designed system is measurable and real. I am particularly keen to deepen my understanding of how large-scale AI pipelines are validated, monitored, and maintained within a safety-critical industrial context. I am confident I would contribute meaningfully from day one while gaining the kind of rigorous engineering discipline that only an organisation of Schlumberger's calibre can provide.

Yours sincerely, Aryan Naik